A Brief Introduction to Experimental Design

BAE 815

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- Originate with a question or problem
- Require a clear articulation of an objective
- Follow a specific plan or procedure (a method)
- Require collection and interpretation of data

- Empirical research consists of:
  - Experimentation
  - Interpretation of results
  - Presentation of results

Philosophy of empirical research

Without proper experimental design/method

- Restricted to mainly exploratory experimentation
- Can gain intuition, but no real answers
- Can be used to help generate hypotheses
- Difficult to justify results to others
- May never appear in published materials

With proper experimental design/method

- Allow full range of types of experimentation
- Can be used to determine conditional answers
- Convince an audience of probable fact
- Facilitates justification of results

Why we need experimental design?

Goals of experimental design

- Validity (Avoid systematic error/bias)
  - Comparison/control
  - Randomization
  - Blocking
  - Blinding
- Reliability (Reduce random error/bias)
  - Replication
  - Balance
- Maximum information from minimum effort/cost

Experimental design is a careful balancing of several features:

- statistical power
- generalizability
- various forms of "validity"
- reliability
- practicality
- cost.

An experiment has treatments, experimental units, and a method to assign treatments to units.

The selection of experimental units and assignment of treatments is called "experiment design."

- Types of variables
- Comparison/control
- Blocking, randomization, and blinding
- Replication
- Factorial experiments

Basic concepts in experimental design

- Explores relationship between variables
  - Independent variable is the variable (factor) that is purposely changed. It is the manipulated variable.
  - Dependent variable changes in response to the independent variable. It is the responding variable.
  - Confounding/nuisance variables are the variables that mask or distort the association between measured variables in a study. They are undesired sources of variation that can affect the dependent variable.

- Constant variables
  - Factors that are kept the same and not allowed to change

- Uncontrollable variables
  - Variables that are known to exist, but conditions prevent them from being manipulated, or it is very difficult to measure them.

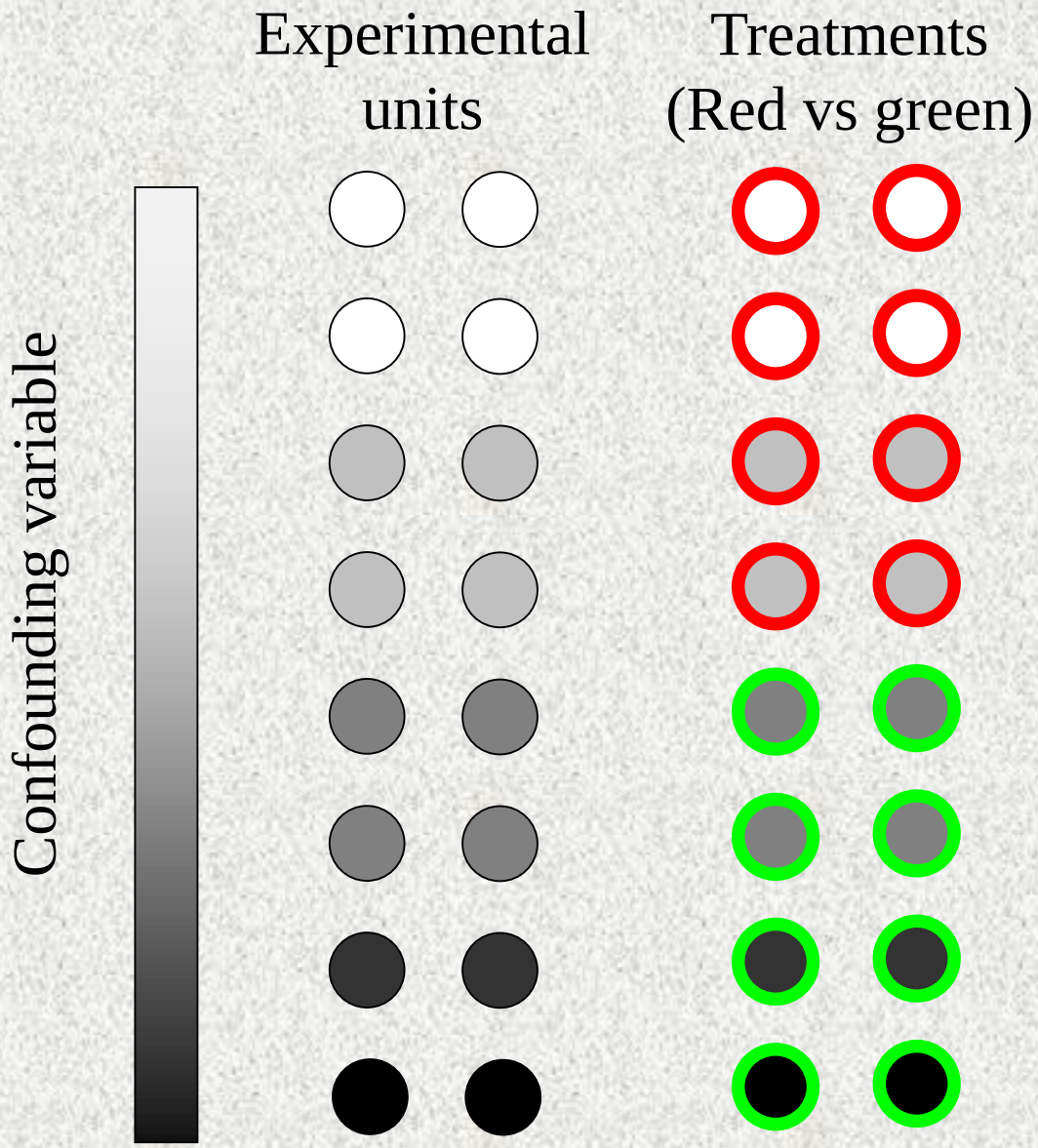
Identify the types of variables

- What is a control?
  - The part of the experiment that serves as the standard of comparison.
- Why is a control necessary?
  - It is the unchanged part of the experiment that detects the effects of hidden variables.
- Internal controls
  - It can be useful to use the subjects themselves as their own controls (e.g., consider the response after vs. before treatment).

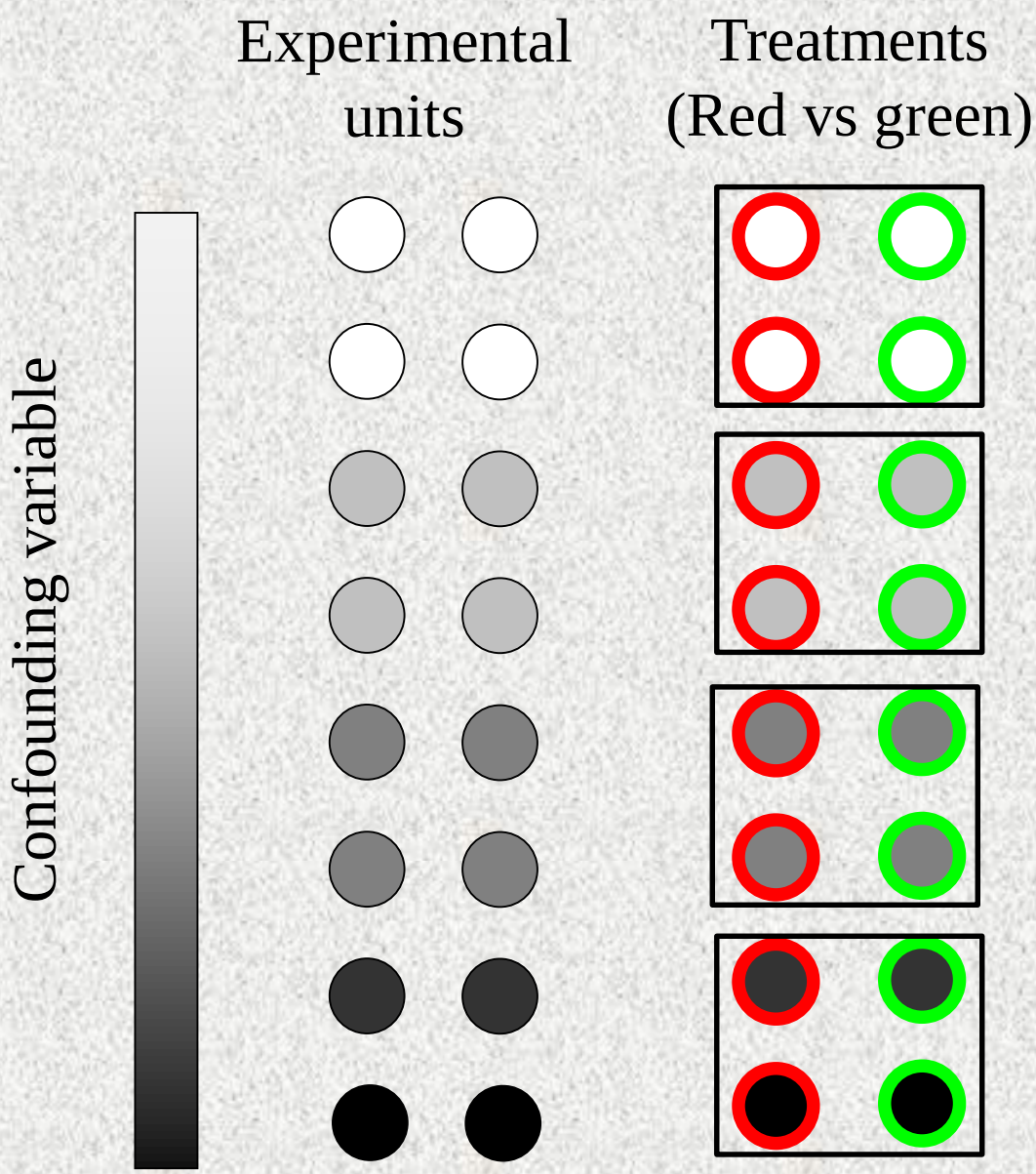
The control in an experiment

- Include the confounding variable as one of the factors in the experiment.
- If you can, fix the confounding variable (make it a constant).
- If you can't fix the confounding variable, use blocking.
- If you can neither fix nor block the confounding variable, use randomization.

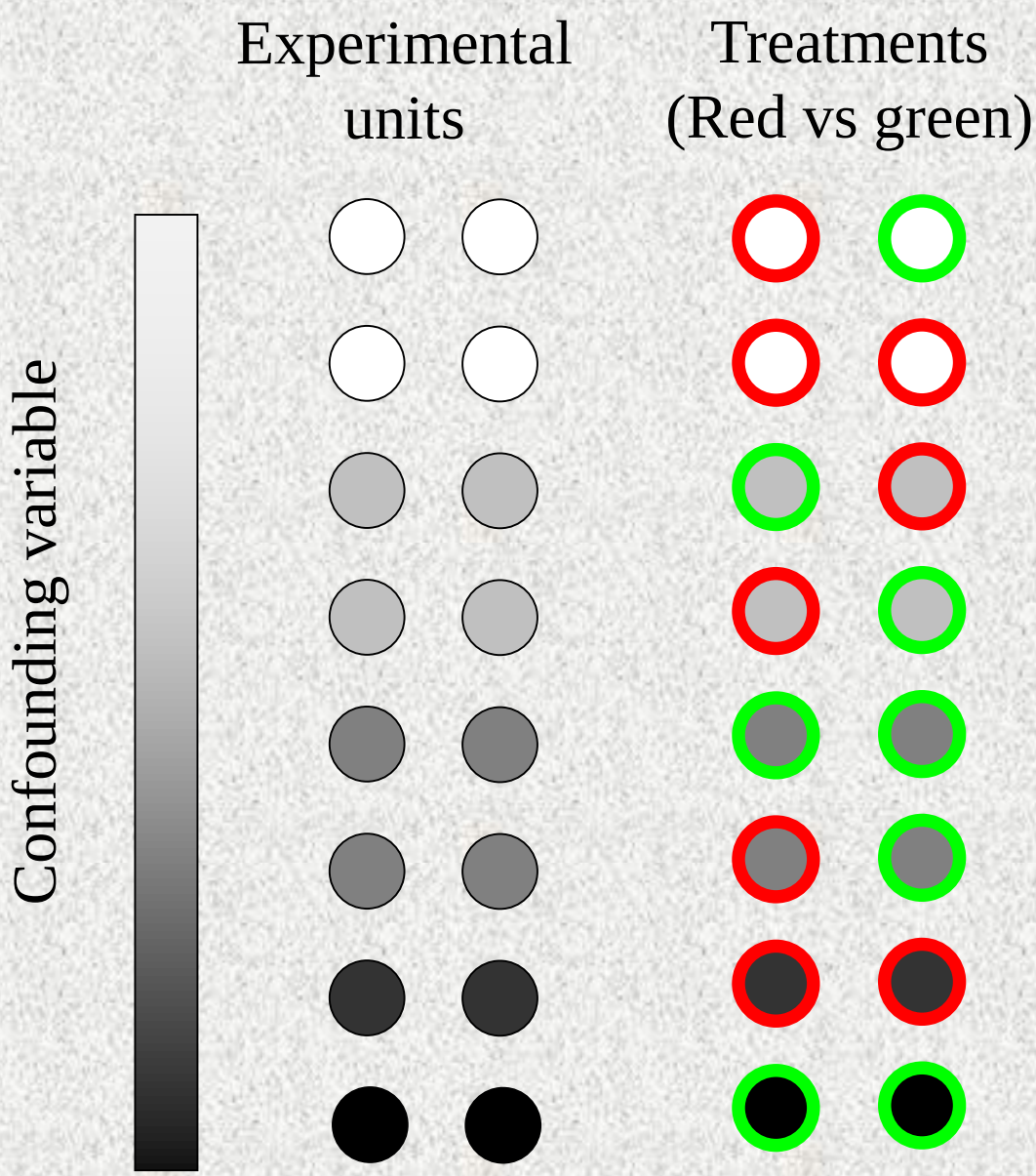
Minimize problems from extraneous sources of variability



Without blocking or randomization, the confounding variable differs among treatments



- A block of units is a set of units that are homogeneous in some sense.
- Blocking is an experimental procedure for isolating variation attributable to a confounding/nuisance variable.



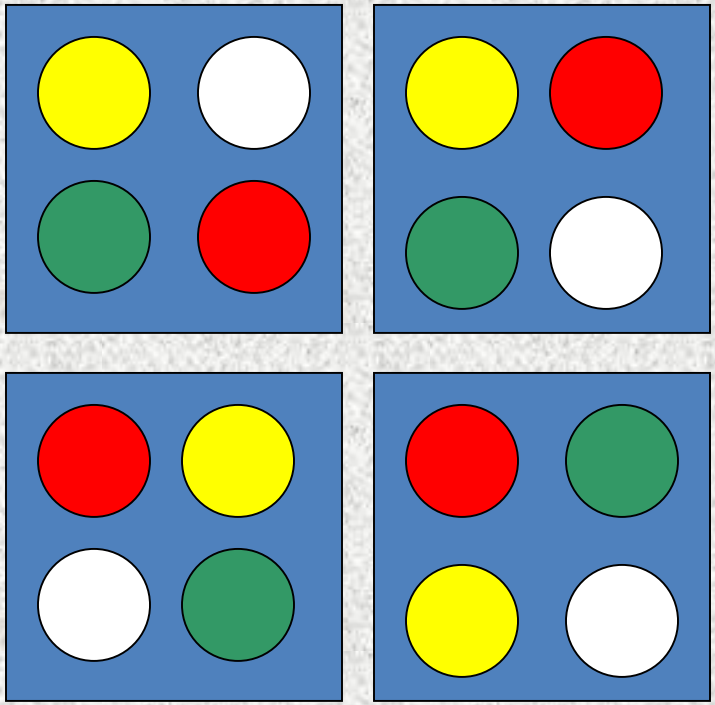
- Randomization is the random assignment of treatments to units.
- With randomization, the confounding variable does not differ among treatments, such that we can assume the differences seen may be largely due to the treatment

- Randomization protects against confounding.
  - Breaks the association between potential confounding variables and the dependent variables
- Randomization can form the basis for inference
  - Randomization allows the later use of probability theory, and so gives a solid foundation for statistical analysis.

Why randomize?

- Are the subjects you are studying really representative of the population you want to study?
- Ideally, your study material is a random sample from the population of interest.

Representativeness and generalizability



Randomized Block Design

	Test session		
	1	2	3
Subject 1	A	B	C
Subject 2	B	C	A
Subject 3	C	A	B

- A Latin Square is a design that blocks for two sources of variation.
- There are two blocking factors in a Latin Square, and these are represented by the rows and columns of the square.
- Each treatment occurs once in each row and once in each column.

Latin Square Design

- Analysis of covariance is another general approach to control confounding/nuisance variables.
- It involves measuring one or more confounding/nuisance variables.
- Through analysis of covariance, the dependent variable can be statistically adjusted to remove the effects of the uncontrolled source of variation.

Analysis of covariance

- Blinding is the concealment of information from the participants and/or researchers about which subjects are receiving which treatments.
- Measurements made by people can be influenced by unconscious biases.
- Two treatments: drug and placebo
  - Single blind: the patients don't know which group they are in, but the doctors do.
  - Double blind: neither the patients nor the doctors administering the drug know which group the patients are in.

Why blinding?

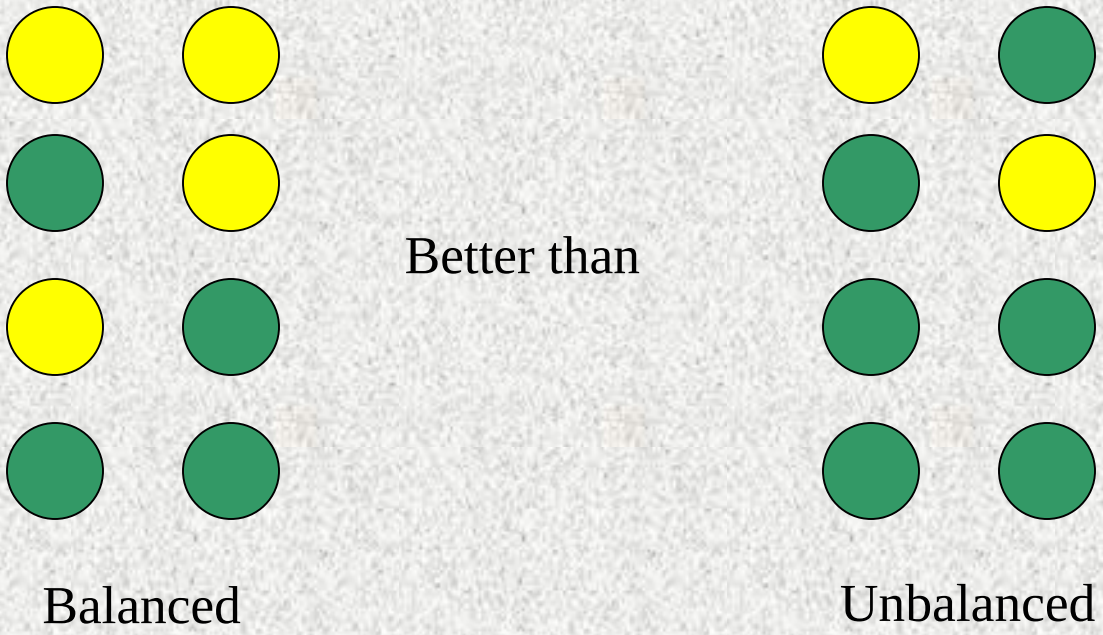
- Reduce the effect of uncontrolled variation (i.e., increase precision). Larger n → Smaller SE

$$SE_{\bar{Y}} = \frac{s}{\sqrt{n}}$$

- Quantify uncertainty.
  - An estimate is of no value without some statement of the uncertainty in the estimate

Why replicate?

- In a balanced experimental design, all treatments have equal sample size
- This maximizes power and makes tests more robust to violating assumptions



Why balance?

- Single factor (one-way):
  - Studies one independent variable
  - Pretest-posttest (one-group)
  - Pretest-posttest (control group)
  - Posttest-only (control group)
- Multi-factor
  - Studies multiple independent variables
  - Each with a number of levels (L)
  - Two-way (e.g., 2 x 2)
  - Three-way (e.g., 2 x 2 x 2)
- Time-series
  - Dependent measure is continuous
  - Establish baseline
  - Measure treatment effect over time

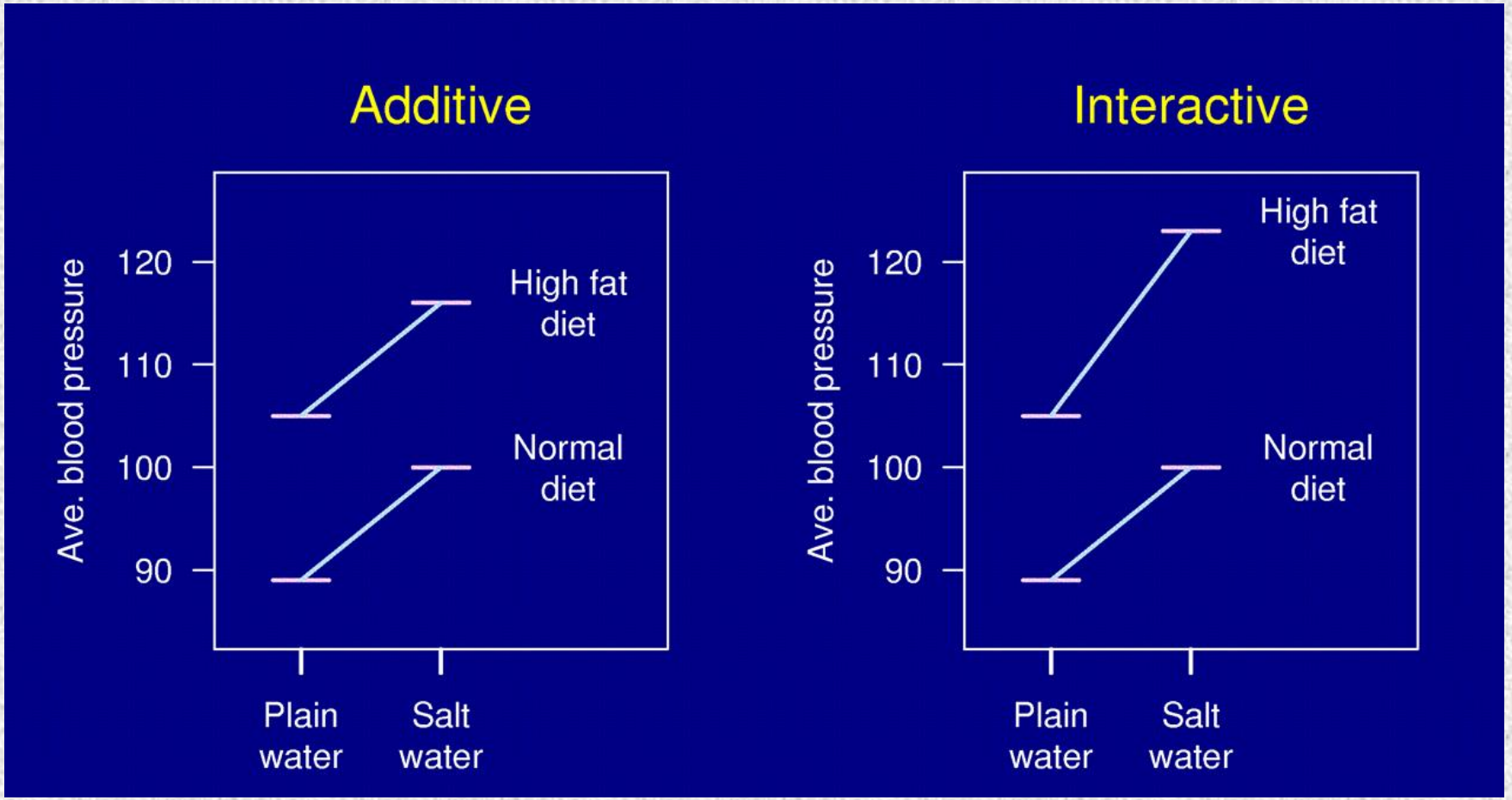
Types of design

Two-way factorial design

- Factorial design
  - Investigates all treatment combinations of two or more variables
  - Permits study of interactions between treatment variables
- Analysis
  - ANOVA

Example: 2 x 3

		Factor2		
		L1	L2	L3
Factor1	L1			
	L2			



Interactions

- Unbiased
  - Randomization
  - Blinding
- High precision
  - Uniform material
  - Replication
  - Stratification/blocking
- Able to estimate uncertainty
  - Replication
  - Randomization

Characteristics of good experiments

- A description of what you used for your experiment, and how you did it.
- Usually should include:
  - Levels of the independent variable
  - Repeated trials
  - Statistics methods

Materials and methods section of your paper

- Data exploration and analysis
- Inductive inference with probability
- Quantification of uncertainty

Why statistics?

- Two possible errors (Type I and Type II)
- Confidence intervals
- Significance level, P-value
- Statistical power
- Determining sample size

Basic concepts in statistics

Use the observed data to answer a yes/no question,

"Does X have an effect on Y?"

- Null hypothesis (H_o): X has no effect on Y.
- Alt. hypothesis (H_a): X does have an effect on Y.

Significance tests

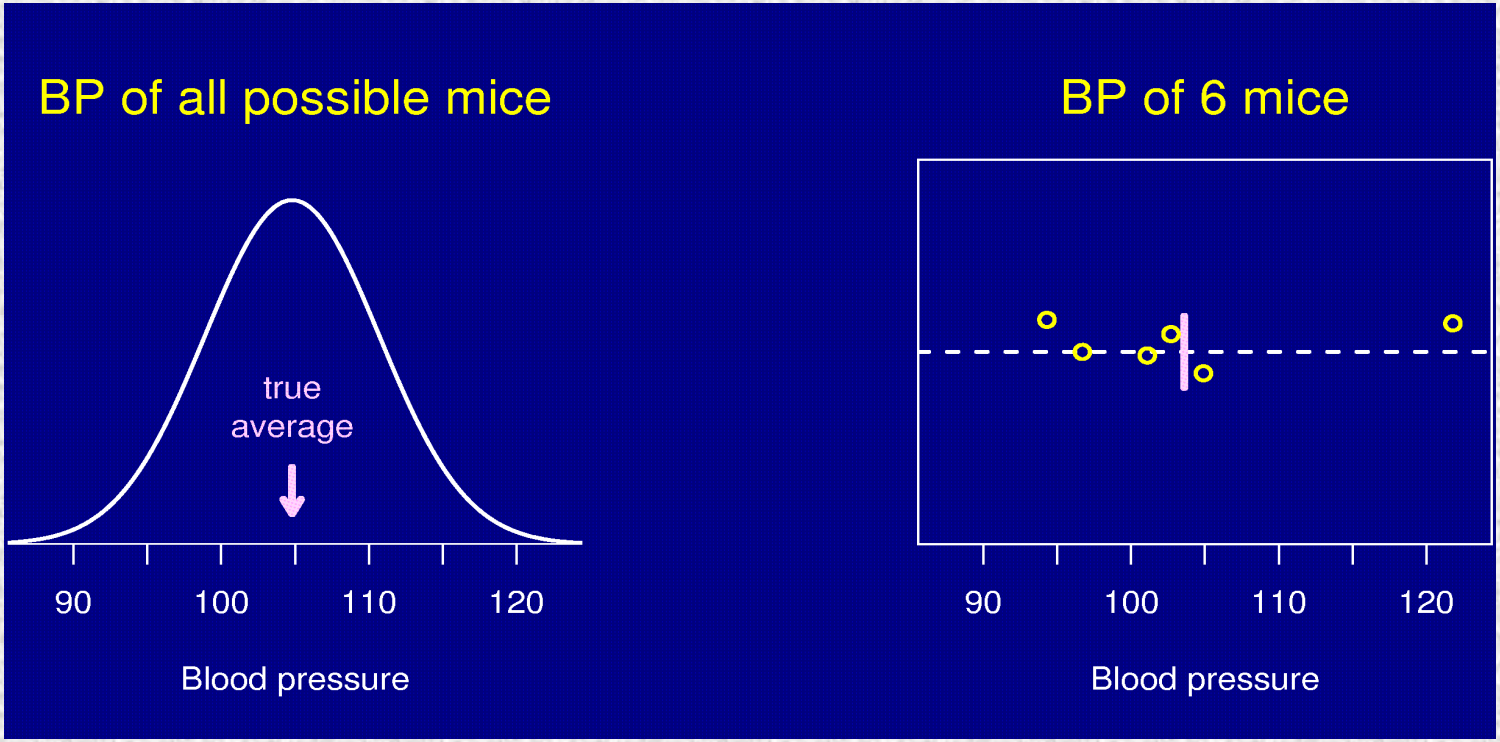
True situation	Our conclusion	
No effect	Not significant	Correct
	Significant (Reject H_0)	Type I error
Has an effect	Significant (Reject H_0)	Correct
	Not significant	Type II error

Type I and Type II Errors

- A false positive result is known as a Type I Error.
- We control for Type I errors explicitly by selecting an appropriate confidence level
- The confidence is the probability of not getting a false positive result.
- Confidence level is the probability of accepting the null hypothesis when the null hypothesis is true.

Confidence level

- Plausible values for the true population average or treatment effect, given the observed data.



95% CI for population average
(93.4 to 113.8)

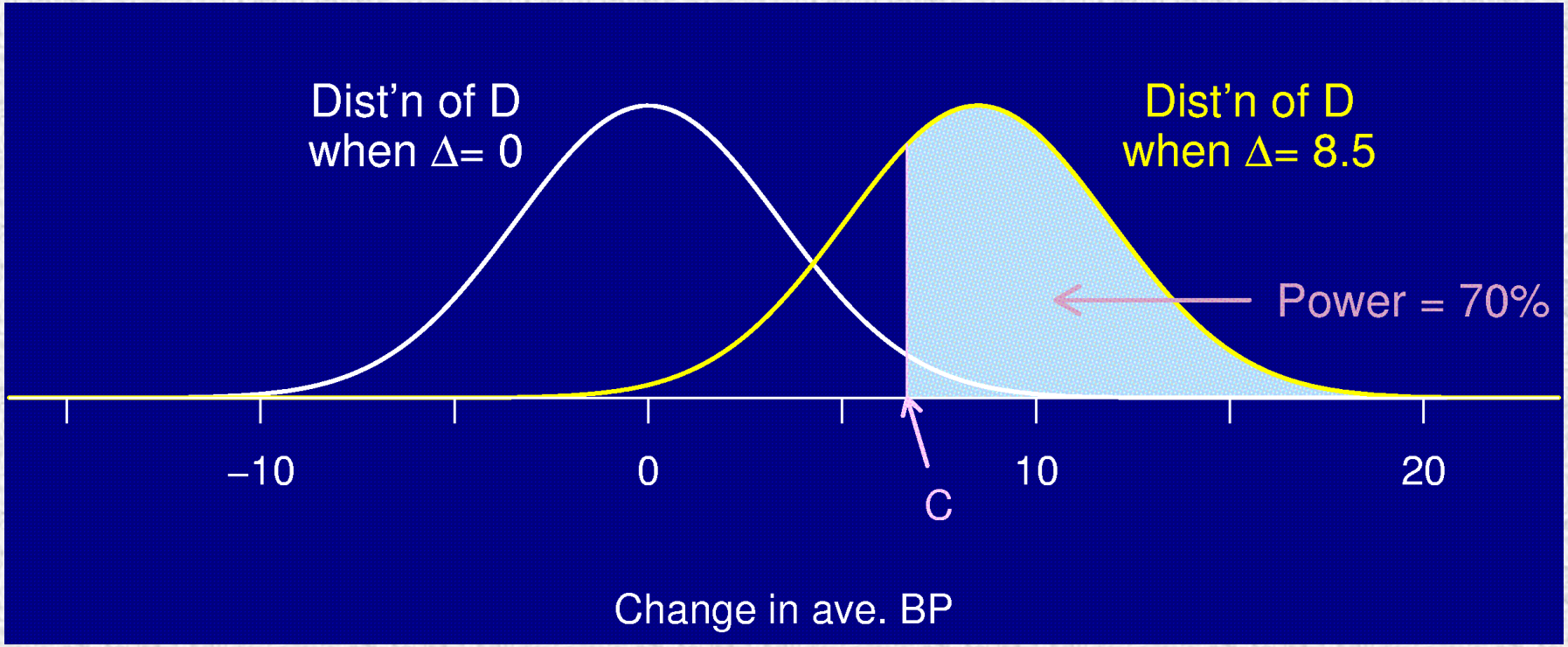
Confidence interval (CI)

- A P-value is the probability of obtaining data as extreme as was observed, if the null hypothesis were true (i.e., if the treatment has no effect).
- If your P-value is smaller than your chosen significance level α, you reject the null hypothesis.
- P-value
  - Summarizes the result of the significance test.
  - Small P-value conclude that there is an effect.

P-values

- A false negative result is known as a Type II Error.
- We control the power implicitly via the confidence level and the experimental design.
- The power is the probability of not getting a false negative result.
- Statistical power is the probability of rejecting the null hypothesis when the null hypothesis is false.

Statistical power



Statistical power

- The structure of the experiment
- The method for analyzing the data
- The size of the true underlying effect
- The variability in the measurements
- The chosen significance level α
- The sample size

Note: We usually try to determine the sample size to give a particular power (e.g. 80%).

Power depends on

- Method for analysis
- Chosen significance level, α (e.g. 5%)
  - More stringent of statistical test (Smaller α) → larger sample size
- Desired statistical power (e.g. 80%)
  - More statistical power → larger sample size
- Variability in the measurements
  - Larger variability → larger sample size
- The smallest meaningful effect
  - Smaller treatment effect → larger sample size

Determining sample size

- Testing the differences between two or more groups on one dependent variable.
  - One way or factorial
- Variation due to treatment and random variation (error)
- Typical breakdown in a between groups design

$$SS_{tot} = SS_{b/t} + SS_e$$

- The F statistic is a ratio of these variances

$$F = MS_b/MS_e$$

Analysis of variances (ANOVA)

- Experiments should be designed.
- Good design and good analysis can lead to reduced sample sizes.
- Consult an expert on both the analysis and the design of your experiment.
- If the results do not make sense, it may suggest a problem in methodology or experimental design

Final words

Sample Data Table

Title: **The Effect of *the independent variable* on the *dependent variable***

Column for independent variable	*Column for dependent variable*			*Column for derived quantity*
Label – with units if necessary	**Label – with units if necessary – multiple trials included**			**Label – with units if necessary. Example = average of trials**
	1	**2**	**3**	
0	**0**	**1**	**0**	**0**
1	**3**	**5**	**4**	**4**
2	**6**	**4**	**7**	**6**
3	**7**	**6**	**8**	**7**
4	**9**	**9**	**8**	**9**

Example

- 20 male mice and 20 female mice.
- Half to be treated; the other half left untreated.
- Can only work with 4 mice per day.

Question: How to assign individuals to treatment groups and to days?

An extremely bad design

Week One					Week Two				
M	Tu	W	Th	F	M	Tu	W	Th	F
C	C	C	C	C	T	T	T	T	T
C	C	C	C	C	T	T	T	T	T
C	C	C	C	C	T	T	T	T	T
C	C	C	C	C	T	T	T	T	T

T = treated, C = control, pink = female, blue = male

Randomized

Week One					Week Two				
M	Tu	W	Th	F	M	Tu	W	Th	F
T	T	T	T	T	C	T	T	C	T
C	T	T	T	T	C	C	C	T	C
C	C	C	T	T	C	C	T	C	C
T	C	C	C	C	C	T	C	T	T

T = treated, C = control, pink = female, blue = male

A stratified/block design

Week One					Week Two				
M	Tu	W	Th	F	M	Tu	W	Th	F
C	T	T	C	T	C	C	T	C	T
T	T	C	C	C	T	T	T	C	C
C	C	T	T	C	C	T	C	T	C
T	C	C	T	T	T	C	C	T	T

T = treated, C = control, pink = female, blue = male